Ziwei Chai

Education

Zhejiang University, Hangzhou, China — 2021.09 – **2026**.06 (expected)
Ph.D in Computer Science, Technology — PhD Advisor: Prof. Yang Yang

Zhejiang University, Hangzhou, China — 09/2017 – 06/2021
B.Eng. in Software Engineering, GPA: 3.95/4, Rank: 1/80

Experience

Zhipu AI GLM Team — 2025.02 – Present
Technical Intern — Advisor: Dr. Xin Lv & Dr. Shulin Cao
Project: **GLM–4** (☆ 6.9k), **GLM–4.5** (☆ 3.1k)
Role: Token efficiency of STEM pre-training data for **GLM–4**, **GLM–4.5** and next-generation model

- High-quality STEM Data Pipeline: Shipped an end-to-end STEM data pipeline that performs filtering, parsing and ranking from the data lake, enabling trillion-token-scale high-quality STEM data assets for GLM-4/4.5 and forthcoming models.
- Synthetic Data Generation: Designed multiple synthetic data generation strategies—including refinement, paraphrasing, and web-based knowledge construction—to alleviate the data wall and deliver consistent pre-training improvements on GLM–4.5.
- Evaluation Framework: Developed a comprehensive evaluation protocol to quantify reasoning knowledge gain of base model during pre-training, designing metrics that more accurately reflect STEM data quality.

Bytedance Seed-Code — 2023.11 – 2025.01
Intern — Advisor: Dr. Guoyin Wang & Prof. Hongxia Yang & Dr. Kai Shen

- RL for Agentic Workflow Orchestration [2]: Leveraged RLVR to train self-orchestrated LLM Code Agents for building multi-agent systems, exploring the scaling laws of multi-agent frameworks.
- Multi-Agent Collaboration System Design [3]: Developed Expert-Token-Routing (ETR), a multi-agent reasoning system using a meta-expert architecture for collaborative reasoning.
- Agentic Data Analysis Benchmark [5]: Implemented InfiAgent-DABench (☆ 134), a comprehensive evaluation framework for benchmarking AI agents on complex data analysis tasks.

Ant Group Alipay — 2022.03 – 2023.03
Research Intern — Advisor: Dr. Weiqiang Wang

- Pioneered a graph neural network-based adaptive money-laundering detection system **[9]**, deployed in Alipay's near-line AML pipeline, achieving a 3% F1 improvement over production baselines.

AINet Lab at Zhejiang University — 2021.09 – Present
Student Researcher — Advisor: Prof. Yang Yang

- LLM for Graph **[6] [7] [8]**, Graph Anomaly Detection **[10]**.

Publications

[1] **GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models** — Technical Report
Ziwei Chai as a contributor to token efficiency of mathematical data pretraining

[2] **AutoSwarm: Training LLM to Self-orchestrate via Reinforcement Learning** — Under Review
Ziwei Chai, Tianjie Zhang, Xuwu Wang, Jing Su, Daoguang Zan, Jie chen, Hanchen Su, Kai Shen, Yang Yang

[3] **An Expert is Worth One Token: Synergizing Multiple Expert LLMs as Generalist via Expert Token Routing** — ACL 2024
Ziwei Chai, Guoyin Wang, Jing Su, Tianjie Zhang, Xuanwen Huang, Xuwu Wang, Jingjing Xu, Jianbo Yuan, Hongxia Yang, Fei Wu and Yang Yang

[4] **Cypher-RI: Reinforcement Learning for Integrating Schema Selection into Cypher Generation** — N
Hanchen Su, Xuyuan Li, Yan Zhou, Zhuoyi Lu, **Ziwei Chai**, Haozheng Wang, Chen Zhang, Yang Yang

[5] **Infiagent-dabench: Evaluating agents on data analysis tasks** ICML 2024
Xueyu Hu, Ziyu Zhao, Shuang Wei, **Ziwei Chai**, Guoyin Wang, Xuwu Wang, Jing Su, Jingjing Xu, Ming Zhu, Yao Cheng and others

[6] **GraphLLM: Boosting graph reasoning ability of large language model** Transactions on Big Data (Cited by 156)
Ziwei Chai, Tianjie Zhang, Liang Wu, Kaiqiao Han, Xiaohai Hu, Xuanwen Huang and Yang Yang

[7] **Enhancing Cross-domain Link Prediction via Evolution Process Modeling** WWW 2025
Xuanwen Huang, Wei Chow, Yize Zhu, Yang Wang, **Ziwei Chai**, Chunping Wang, Lei Chen, Yang Yang

[8] **Can GNN be Good Adapter for LLMs?** WWW 2024
Xuanwen Huang, Kaiqiao Han, Yang Yang, Dezheng Bao, Quanjin Tao, **Ziwei Chai** and Qi Zhu

[9] **Towards Learning to Discover Money Laundering Sub-network in Massive Transaction Network** AAAI 2023
Ziwei Chai, Yang Yang, Jiawang Dan, Sheng Tian, Changhua Meng, Weiqiang Wang and Yifei Sun

[10] **Can abnormality be detected by graph neural networks?** IJCAI 2022
Ziwei Chai, Siqi You, Yang Yang, Shiliang Pu, Jiarong Xu, Haoyang Cai and Weihao Jiang

[11] **Time2Graph+: Bridging time series and graph representation learning via multiple attentions** TKDE 2021
Ziwei Chai, Siqi You, Yang Yang, Shiliang Pu, Jiarong Xu, Haoyang Cai and Weihao Jiang

Selected Awards

- Outstanding Graduate of Zhejiang Province 2021
- Chinese National Scholarship (top 1%) 2019